HARSHITH VH

Masters in Physics
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Education

Degree	Institution	Year	Score
BSc Physics & Computer Science	Kristu Jayanti (Deemed to be University)	2023–2026	8.84 / 10
XI – XII (Science)	The National Pre University College	2021–2023	87%

Internship

Research Internship – Cosmic Ray Laboratory, TIFR (NCRA), Ooty | June 2025
- Worked with the Muon Telescope setup at the Cosmic Ray Laboratory, Ooty, to study secondary cosmic ray muon flux at ground level.
- Operated and calibrated scintillator-based detectors used in the muon telescope array for particle detection and event triggering.
- Performed data acquisition, quality checks, and preliminary analysis of proportional counter data.
- Gained hands-on experience in high-energy particle detection instrumentation and data analysis pipelines in an experimental astrophysics environment.

Work Experience

Physics Tutor – Himalaya Tuitions | May 2026 – Present
- Deliver engaging, face-to-face physics instruction, utilizing classroom whiteboards and physical demonstrations to break down complex mathematical derivations.
- Foster an interactive physical learning environment that encourages real-time questions, peer collaboration, and active participation.

Research Interests

Field	Topics
Gravitational Waves & Data Analysis	Compact Binary Objects, Q-transform Analysis, Wavelet Transforms, Time-Frequency Analysis, Bayesian Inference for Source Parameter Estimation

AI & Machine Learning	Supervised and Unsupervised Algorithms, Deep Learning, Neural Networks, Large Language Models (LLMs), Retrieval- Augmented Generation (RAG)
Exoplanet Science	Transit Curve Analysis, High-Contrast Imaging, PSF Subtraction (KLIP, PyKLIP), Flux Calibration & Planet Mass Estimation
Cosmic Ray Physics	Muon Detection, Scintillator-based Detectors, Data Acquisition & Calibration

Academic Coursework

- Classical Mechanics
- Quantum Mechanics
- Mathematical Physics
- Analog and Digital Electronic
- Semiconductor Physics
- Condensed Matter Physics
- Atomic, Molecular and Nuclear Physics
- Mechanics and Properties of Matter
- Optics
- Heat and Thermodynamics
- Electricity and Magnetism
- Database Management System and SQL.
- Programming in C
- Data Structures and Algorithms
- Java Programming
- Internet Technology (HTML, CSS, JavaScript)
- PHP Programming
- Software Engineering
- Scientific Computing using Python – IIT Kanpur via NPTEL
- Applications of Modern Physics

Research & Projects

Evolving Sensitivity of LIGO Livingston: Inspiral Range and Mass Scaling Across Observing Runs [Research] | 2025

- Analyzed LIGO Livingston detector sensitivity across O1, O2, and O3 using Amplitude Spectral Density (ASD) and noise characterization techniques.
- Computed inspiral ranges for BNS, BBH, and BHNS systems, demonstrating significant sensitivity improvement (e.g., BNS: ~63 Mpc → ~110 Mpc).
- Investigated chirp mass scaling relation, identifying deviation from theoretical d∝M^(5/6) with an observed power law d∝ M^0.67.
- Developed a Python-based data analysis pipeline using GWOSC data for ASD computation, interpolation, and numerical integration.
- Provided insights into detector noise reduction, mass-dependent sensitivity, and implications for future runs (O4).

Q-Transform Analysis of GW230529 [Research] | 2024

- Performed Q-transform time-frequency analysis of GW230529, a suspected neutron star–black hole (NSBH) merger.

- Identified a significant signal between −9 and −9.2 s relative to trigger time, suggesting potential early inspiral or noise artefacts.
- Demonstrated the effectiveness of time-frequency methods in gravitational wave detection and interpretation.

GW-Interpreter: A Physics-Informed Large Language Model Pipeline for Automated Astrophysical Interpretation of Gravitational Wave Events | 2026

- Developed a physics informed LLM pipeline to automatically interpret gw event parameters from public GWOSC catalogs.
- Designed a evaluation framework based on factual accuracy, physics concept coverage and reasoning depth to benchmark interpretation quality against generic LLM outputs.
- Demonstrated improved scientific reasoning and domain specific physical interpretation using prompt engineering.

CMB Foreground Separation using PCA & ICA | 2026

- This project stimulates the realistic multi frequency flat sky CMB observation and applies two ML component searching algorithms PCA & ICA to recover the primordial CMB map.

Classification of Gravitational Wave Events Based on Sources | 2024

- Developing a machine learning model to classify GW events into BBH, BNS, and NSBH mergers using astrophysical parameters extracted from GWOSC data.
- Focused on improving event classification accuracy through feature engineering and model comparison.

Predicting Habitability of Exoplanets | 2024

- Applied Logistic Regression to classify exoplanet habitability using planet radius, mass, equilibrium temperature, stellar mass, and other features.
- Model predicts whether an exoplanet falls within the habitable zone.

Earthquake Magnitude Prediction Using Machine Learning | 2025

- Developed and optimised Random Forest, Gradient Boosting, and SVR models on historical Indian seismic data.
- Analysed model performance using statistical metrics and visualised prediction accuracy.

Rainfall Classification Using Machine Learning |2025

- Implemented Random Forest, SVM, and Logistic Regression to classify annual rainfall patterns in India into Low, Medium, and High categories.
- Applied feature engineering techniques on historical weather data to enhance model performance.

Ionic Plasma Thruster – Experimental Model | Kristu Jayanti College | 2025

- Developed a working prototype of an ionic plasma thruster that accelerates ionised particles via a high-voltage electric field.
- Explored feasibility for satellite propulsion and deep-space mission applications, comparing efficiency against conventional chemical rockets.

Campus Sphere – Full-Stack Web Application | 2025

- Building a full-stack campus social and resource platform featuring event discovery, club management, academic resource sharing, and student networking.
- Designed a scalable backend architecture with a responsive UI, focusing on seamless user experience for college communities.

Conference Presentation

1. *Harshith V. H.*, Vaishnavi K., B. Hariharan (Tata Institute of Fundamental Research), et al.
   “Characteristic Analysis of Proportional Counters and Plastic Scintillators Used in the GRAPES-3 Experiment”
   National Conference [NCARAS-2025], *MVJ College of Engineering*, Bengaluru
   (Focus: detector characterization, cosmic ray instrumentation, GRAPES-3 experiment)

Workshops [Conducted]

Title	Event / Venue	Year
Introduction to Gravitational Waves and Data Analysis	Student Enrichment Program for BSc Physics Students	2026
Exoplanet: Transit Curve Analysis and Advanced Imaging	Student Enrichment Program for BSc Physics Students	2024
Hands-On Session on Transit Curve Analysis	Student Enrichment Program for BSc Physics Students	2024
Cosmic Conundrums – Host	National Space Day	2024

Workshops & Schools [Attended]

Title	Institution	Year
ISRO START Program 2026	Indian Space Research Organisation (ISRO)	2026
LIGO Undergraduate Study Hub	California Institute of Technology	2025
NASA Open Science 101	NASA	2025
PCB Designing Workshop	Altium	2025
ICTS-TIFR Gravitational Waves Open Data Workshop	ICTS, Bangalore	2024
NASA Sagan Workshop	California Institute of Technology	2024
Pravega Astrophysics Workshop	IISc, Bangalore	2023

Certifications

- ISRO Start Program – ISRO (2026)
- Deep Learning with TensorFlow – IBM (2026)
- Data Science 101 – IBM (2026)
- Operating Systems – NPTEL (2025)
- Applications of Modern Physics – KJC (2024)
- Python Programming – Infosys Springboard (2024)

Awards & Honours

Award	Institution / Event	Year
First Prize – Poster Presentation, National Conference NCARAS-2025	MVJ College of Engineering (Autonomous), Bengaluru	2025
Synergia – Physics Fest Winner	Kristu Jayanti College	2024
Short Film Making – Winner	Kristu Jayanti College	2023
Merit – XII	The National Pre University College	2023

Technical Skills

- **Programming:** Python, C, C++, Julia, Java
- **Machine Learning:** Scikit-Learn, TensorFlow, XGBoost, Random Forest, SVM, PCA
- **AI:** LLMs, NLP, RAG, Prompt Engineering
- **Gravitational Wave Analysis:** GWpy, PyCBC, Bilby, Q-transform, Bayesian Inference
- **Exoplanet Science:** PyKLIP, PSF Subtraction, Flux Calibration, Transit Curve Analysis
- **Web Development:** HTML, CSS, JavaScript, PHP, MySQL, React
- **Tools & Software:** LaTeX, Jupyter Notebook, Git, Linux (Ubuntu)
- **Data Analysis & Visualisation:** Pandas, NumPy, Matplotlib, Seaborn, Scipy

Last updated: September 2026